

INSTRUCTIONS TO CANDIDATES

Please read these instructions carefully, but **do not open this question paper until you are ready to begin.**

This paper contains **27 multiple-choice questions**. There are no penalties for incorrect responses, only marks for correct answers, so you should attempt all of the questions. Each question is worth one mark.

For each question, choose the one option you consider correct. If you make a mistake, erase thoroughly and try again.

You must complete the paper within the time limit.

You can use the question paper for rough working. Dictionaries and calculators are **NOT permitted**.

Please wait until you are ready before turning this page.

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Chemistry

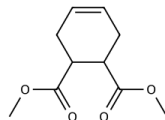
- 1 A sample of element ${}^{\text{X}}\text{X}$ consists of only three isotopes: ${}^{68}\text{X}$, ${}^{70}\text{X}$, and ${}^{72}\text{X}$. The relative atomic mass (A_r) of ${}^{\text{X}}\text{X}$ in this sample is 70.2.

isotope	percentage abundance / %
${}^{68}\text{X}$	x
${}^{70}\text{X}$	30
${}^{72}\text{X}$	y

What is the value of y ?

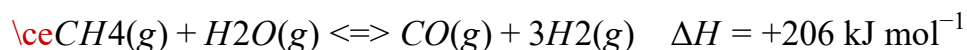
- A 20
B 30
C 35
D 40
E 45
- 2 Magnesium is an element in Group 2 of the Periodic Table.
Which of the following statements is/are correct about magnesium?
- 1 Magnesium is a stronger reducing agent than calcium.
2 Magnesium has a smaller atomic radius than calcium.
3 Magnesium reacts with nitrogen gas to form a compound containing 72% magnesium by mass.
- (A_r values: ${}^{\text{N}}\text{N} = 14$; ${}^{\text{Mg}}\text{Mg} = 24$; ${}^{\text{Ca}}\text{Ca} = 40$)
- A none of them
B 1 only
C 2 only
D 3 only
E 1 and 2 only
F 1 and 3 only
G 2 and 3 only
H 1, 2 and 3

- 3 The skeletal formula of an organic compound is shown below.
What is the empirical formula of this compound?



- A $\text{C}_5\text{H}_7\text{O}_2$
- B $\text{C}_{10}\text{H}_{14}\text{O}_4$
- C $\text{C}_5\text{H}_6\text{O}_2$
- D $\text{C}_5\text{H}_8\text{O}_2$
- E $\text{C}_4\text{H}_7\text{O}_2$
- F $\text{C}_2\text{H}_3\text{O}$

- 4 The industrial production of synthesis gas from methane is represented by the following endothermic, reversible reaction:



A mixture of $\text{CH}_4(\text{g})$ and $\text{H}_2\text{O}(\text{g})$ is placed in a closed vessel and allowed to reach dynamic equilibrium.

Which of the following statements is/are correct?

- 1 At dynamic equilibrium, the activation energy of the forward reaction is equal to the activation energy of the backward reaction.
- 2 Increasing the temperature at constant pressure increases the rate of both the forward and backward reactions, and increases the equilibrium yield of $\text{H}_2(\text{g})$.
- 3 Adding a catalyst decreases the time taken to reach equilibrium, but does not change the ratio $\frac{k_{\text{forward}}}{k_{\text{backward}}}$, where k_{forward} and k_{backward} are the rate constants for the forward and backward reactions.

- A none of them
- B 1 only
- C 2 only
- D 3 only
- E 1 and 2 only
- F 1 and 3 only
- G 2 and 3 only
- H 1, 2 and 3

- 5 Two experiments were carried out using separate samples of an aqueous solution of an unknown base B of concentration $0.150 \text{ mol dm}^{-3}$ (11.1 g dm^{-3}), and the following observations were made:

* 20.0 cm^3 of the base solution was completely neutralised by 30.0 cm^3 of $0.200 \text{ mol dm}^{-3}$ hydrochloric acid, HCl(aq) .

* A pale blue precipitate formed when a few drops of aqueous copper(II) sulfate, $\text{CuSO}_4(\text{aq})$, were added to a portion of the base solution.

Which of the following statements can be deduced from this information?

1. One mole of the base reacts with two moles of H^+ ions in complete neutralisation.
2. The base is fully dissociated in aqueous solution.
3. The relative formula mass (M_r) of the base is 74.

(A_r values are not required)

- A none of them
- B 1 only
- C 2 only
- D 3 only
- E 1 and 2 only
- F 1 and 3 only
- G 2 and 3 only
- H 1, 2 and 3

- 6 What are the oxidation states of sulfur in SO_2 and SO_4^{2-} , respectively?

- A +2 and +4
- B +4 and +6
- C +4 and +4
- D +6 and +4
- E -4 and -6
- F 0 and +6

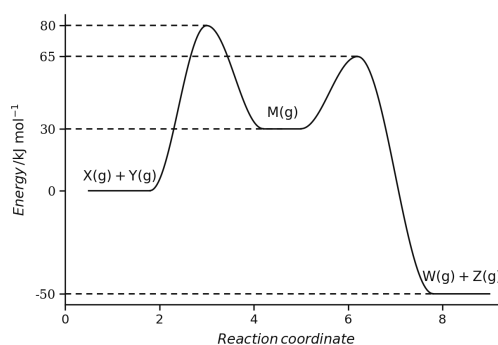
- 7 The diagram shows the reaction energy profile for the two-step gas-phase reaction:



where M is a reaction intermediate.

Which of the following statements is/are correct?

- 1 The rate-determining step for the forward reaction is the conversion of intermediate M into the products $\text{W} + \text{Z}$.
- 2 The activation energy for the overall reverse reaction is $+130 \text{ kJ mol}^{-1}$.
- 3 The conversion of intermediate M back into reactants has a higher activation energy than its conversion into products.

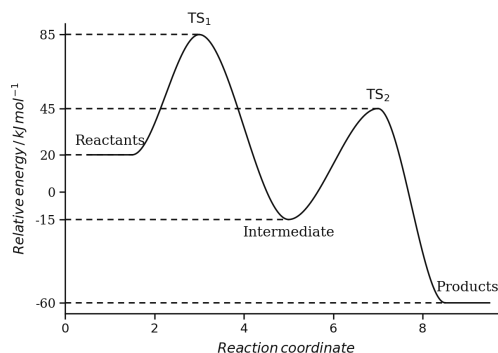


- A none of them
- B 1 only
- C 2 only
- D 3 only
- E 1 and 2 only
- F 1 and 3 only
- G 2 and 3 only
- H 1, 2 and 3

- 8 The energy profile for a two-step reaction, $\text{Reactants} \rightarrow \text{Intermediate} \rightarrow \text{Products}$, is shown in the diagram.

Which of the following statements is/are correct?

- 1 The overall reaction is exothermic with $\Delta H = -80 \text{ kJ mol}^{-1}$.
- 2 The first step ($\text{Reactants} \rightarrow \text{Intermediate}$) is the rate-determining step for the forward reaction.
- 3 The activation energy for the overall reverse reaction ($\text{Products} \rightarrow \text{Reactants}$) is 105 kJ mol^{-1} .



- A none of them
- B 1 only
- C 2 only
- D 3 only
- E 1 and 2 only
- F 1 and 3 only
- G 2 and 3 only
- H 1, 2 and 3

- 9 A diol reacts with an excess of propanoic acid in the presence of an acid catalyst to form a diester.

The diester has a relative molecular mass (M_r) of 174.

What is the relative molecular mass (M_r) of the diol?

(A_r values: $H = 1.0$; $C = 12.0$; $O = 16.0$)

- A 26
B 44
C 62
D 76
E 90
F 100
G 118
H 136
- 10 Researchers intend to isolate mitochondria from a liver cell homogenate using differential centrifugation. The table below displays the minimum gravitational force (g) required to pellet specific cellular components.

Component	Minimum Force to Pellet (g)
Nuclei	1,000
Mitochondria	15,000
Ribosomes	100,000

The homogenate is first centrifuged at 1,000 g for 10 minutes, producing a pellet and a supernatant.

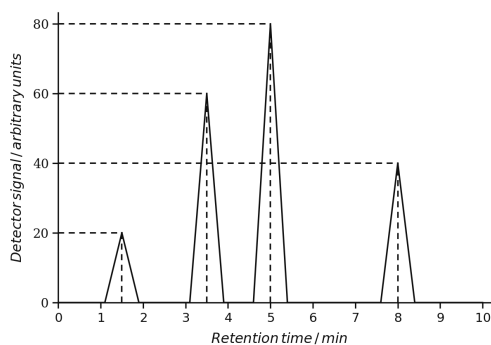
Which of the following subsequent steps would yield the purest sample of mitochondria?

- A Resuspend the pellet from the first step and centrifuge at 15 000 g.
B Centrifuge the supernatant from the first step at 15 000 g and collect the resulting pellet.
C Centrifuge the supernatant from the first step at 15 000 g and collect the resulting supernatant.
D Centrifuge the supernatant from the first step at 100 000 g and collect the resulting pellet.
E Centrifuge the supernatant from the first step at 100 000 g and collect the resulting supernatant.

- 11 A mixture containing four alkanes—two isomers with molecular formula C_5H_{12} (pentane and 2-methylbutane) and two alkanes with molecular formula C_6H_{14} (hexane and 2,2-dimethylbutane)—was analysed using gas-liquid chromatography with a non-polar stationary phase and helium carrier gas.

The resulting chromatogram is shown below. All four peaks have the same base width, and the detector response factor is identical for all four compounds.

What is the percentage by mass in the mixture of the C_5H_{12} isomer that has the strongest attraction to the stationary phase relative to the mobile phase?

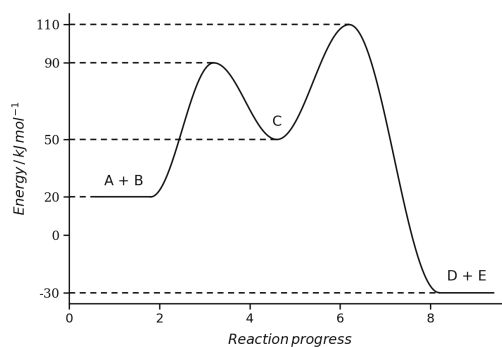


- A 10%
- B 20%
- C 30%
- D 40%
- E 60%
- F 75%

- 12 The reaction energy profile for the two-step reversible reaction $A + B \rightleftharpoons C \rightleftharpoons D + E$ is shown in the diagram.

Which row in the table gives the activation energy of the rate-determining step for the reverse reaction ($D + E \rightarrow A + B$) and the overall enthalpy change (ΔH) of the forward reaction?

	Activation energy of rate-determining step for reverse reaction / kJ mol^{-1}	Overall enthalpy change (ΔH) of forward reaction / kJ mol^{-1}
A	+40	-50
B	+60	+50
C	+90	-50
D	+140	-50
E	+140	+50
F	+160	-50

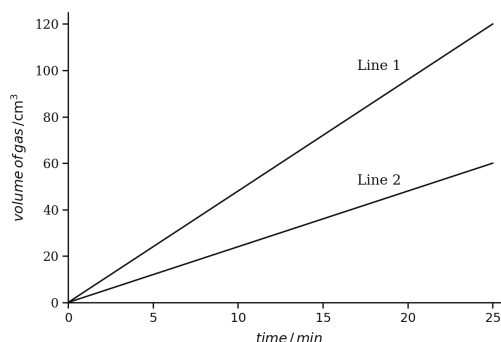


- 13** An aqueous solution of dilute sulfuric acid, $\text{H}_2\text{SO}_4(\text{aq})$, is electrolysed at room temperature and pressure using inert platinum electrodes and a constant electric current.

The graph shows the volume of gas collected at each of the two electrodes as a function of time.

Which of the following statements is/are correct?

1. Line 1 represents the gas collected at the cathode.
2. In the external circuit, electrons flow from the electrode producing the gas of Line 1 towards the electrode producing the gas of Line 2.
3. Doubling the electric current would double the gradient of both Line 1 and Line 2.



- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

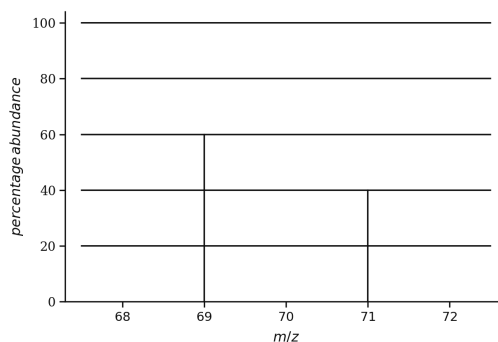
- 14 The mass spectrum of a sample of element M is shown in the diagram.

A 0.100 mol sample of M reacts completely with 2.40 g of oxygen gas (O_2) to form a single oxide.

Which row in the table gives the relative atomic mass of this sample of M and the empirical formula of the oxide formed?

($A_r(O) = 16.0$)

	Relative atomic mass	Empirical formula of oxide
A	69.8	MO
B	69.8	M_2O_3
C	69.8	MO_2
D	70.0	M_2O_3
E	70.0	MO_2
F	70.2	MO
G	70.2	M_2O_3
H	70.2	MO_2



15 Which of the following statements is/are correct?

1. Ethene decolourises bromine water.
2. Addition polymerisation of ethene produces a small-molecule by-product.
3. Ethanol reacts with sodium to release hydrogen gas.

- A** none of them
B 1 only
C 2 only
D 3 only
E 1 and 2 only
F 1 and 3 only
G 2 and 3 only
H 1, 2 and 3

16 A solution contains both hydrochloric acid, $\text{HCl}(\text{aq})$, and sulfuric acid, $\text{H}_2\text{SO}_4(\text{aq})$.

- A 50.0 cm^3 sample of this solution is completely neutralized by 35.0 cm^3 of $0.200 \text{ mol dm}^{-3}$ $\text{NaOH}(\text{aq})$.

- A separate 50.0 cm^3 sample of the same solution is treated with excess aqueous barium chloride, $\text{BaCl}_2(\text{aq})$, producing 0.466 g of dry $\text{BaSO}_4(\text{s})$ precipitate.

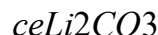
What is the concentration of hydrochloric acid in the original solution?

(A_r values: $\text{H} = 1.0$; $\text{O} = 16.0$; $\text{S} = 32.0$; $\text{Ba} = 137.0$)

- A** $0.020 \text{ mol dm}^{-3}$
B $0.030 \text{ mol dm}^{-3}$
C $0.040 \text{ mol dm}^{-3}$
D $0.060 \text{ mol dm}^{-3}$
E $0.080 \text{ mol dm}^{-3}$
F $0.100 \text{ mol dm}^{-3}$

- 17 Three separate experiments are carried out. In each experiment, an excess of dilute hydrochloric acid is reacted with 0.050 mol of a solid Group 1 carbonate:

- Experiment 1:



- Experiment 2:



- Experiment 3:



Let V_1 , V_2 , and V_3 be the final volumes of gas produced in Experiments 1, 2, and 3 respectively, measured at room temperature and pressure.

Which of the following correctly compares these volumes?

A $V_1 < V_2 < V_3$

B $V_1 > V_2 > V_3$

C $V_1 = V_2 = V_3$

D $V_1 = V_2 < V_3$

E $V_1 > V_2 = V_3$

- 18 The ions X^{2-} , Y^- and Z^{2+} are isoelectronic (they have the same electronic configuration).

Which of the following statements about the **neutral atoms** X, Y and Z is correct?

A Element Z has the lowest atomic number.

B Element Z is in the period immediately following elements X and Y.

C Elements X, Y and Z are in the same period.

D Element X is in Group 2 of the Periodic Table.

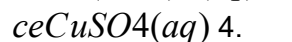
E Element Y has a larger atomic radius than element Z.

- 19 Electrolysis is carried out on separate samples of the following four aqueous solutions using inert graphite electrodes:

1.



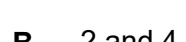
2.



3.



4.



In which of these experiments is the metal deposited at the cathode?

A 1 and 3 only

B 2 and 4 only

C 1, 2 and 3 only

D 3 and 4 only

E 1, 2, 3 and 4

- 20** The boiling points of the following three compounds are -100°C , -24°C , and 13°C (not necessarily in that order): 1. Boron trifluoride (BF_3) 2. Chloromethane (CH_3Cl) 3. Boron trichloride (BCl_3)

Which option lists the compounds in order of **increasing** boiling point? (Ar values: H = 1; B = 11; C = 12; F = 19; Cl = 35.5)

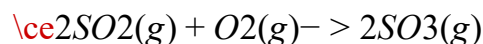
- A** $\text{BF}_3 < \text{CH}_3\text{Cl} < \text{BCl}_3$
B $\text{BF}_3 < \text{BCl}_3 < \text{CH}_3\text{Cl}$
C $\text{CH}_3\text{Cl} < \text{BF}_3 < \text{BCl}_3$
D $\text{BCl}_3 < \text{CH}_3\text{Cl} < \text{BF}_3$

- 21** An element, M, is in Period 3 of the Periodic Table. The first four successive ionisation energies of M are 578 kJ mol^{-1} , 1817 kJ mol^{-1} , 2745 kJ mol^{-1} , and $11578 \text{ kJ mol}^{-1}$.

Which row in the table correctly identifies the formula of the oxide of M, the acid-base character of this oxide, and whether this oxide reacts with excess aqueous sodium hydroxide?

	formula of oxide	acid-base character of oxide	reaction with excess NaOH(aq)
A	M_2O_3	amphoteric	reacts
B	M_2O_3	basic	does not react
C	M_2O_3	basic	reacts
D	M_2O_3	acidic	reacts
E	MO	basic	does not react
F	MO	amphoteric	reacts
G	MO_2	acidic	reacts
H	MO_2	amphoteric	reacts

- 22 The reaction energy profile for the oxidation of sulfur dioxide is shown below:

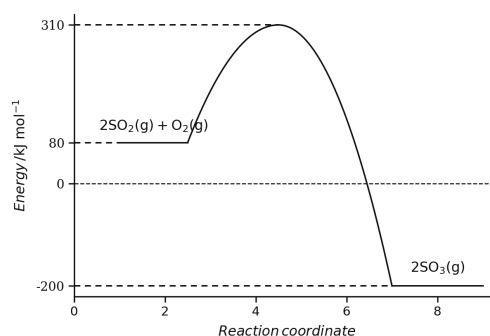


In an experiment, the thermal energy released by the complete oxidation of a sample of $\text{SO}_2(\text{g})$ is transferred to 600 g of water. The heat transfer process is 75% efficient.

The temperature of the water increases by 25°C .

What mass of SO_2 was reacted?

(A_r values: S = 32.0, O = 16.0. Specific heat capacity of water = $4.2 \text{ J g}^{-1}\text{C}^{-1}$)



- A 10.8 g
B 19.2 g
C 21.6 g
D 38.4 g
E 48.0 g
F 53.8 g
- 23 The single-step gas-phase reaction $\text{X}(\text{g}) + 3\text{Y}(\text{g}) \rightleftharpoons 2\text{Z}(\text{g})$ is at equilibrium in a sealed container. The volume of the container is suddenly halved while the temperature is maintained at a constant value. What is the value of the ratio below, immediately after the change?

$$\frac{\text{rate of forward reaction}}{\text{rate of reverse reaction}}$$

- A $\frac{1}{4}$
B $\frac{1}{2}$
C 1
D 2
E 4
F 16

- 24 The structure of propan-2-ol is shown below:

Propanone reacts with hydrogen gas to form propan-2-ol according to the following equation:



Using the mean bond enthalpies provided below, what is the mean bond enthalpy of the $\text{C} = \text{O}$ bond in propanone?

Mean bond enthalpies:

- $\text{C} - \text{C}$: 348 kJ mol^{-1}
- $\text{C} - \text{H}$: 413 kJ mol^{-1}
- $\text{C} - \text{O}$: 358 kJ mol^{-1}
- $\text{H} - \text{H}$: 436 kJ mol^{-1}
- $\text{O} - \text{H}$: 464 kJ mol^{-1}



- A 332 kJ mol^{-1}
 B 440 kJ mol^{-1}
 C 745 kJ mol^{-1}
 D 853 kJ mol^{-1}
 E 1181 kJ mol^{-1}
- 25 A chemical equation that represents the reaction of copper(I) phosphide with dilute nitric acid is:



where w , x , y , and z are integers.

What is the value of the sum $w + x + y + z$?

- A 49
 B 53
 C 57
 D 59
 E 63

- 26** When white phosphorus, P_4 , reacts with warm aqueous sodium hydroxide, it undergoes a disproportionation reaction to produce phosphine gas, PH_3 , and hypophosphite ions, H_2PO_2^- .

The unbalanced ionic equation for this reaction is:



What is the simplest whole-number ratio of $e : d$ in the balanced equation?

- A** 1 : 1
 - B** 1 : 2
 - C** 2 : 1
 - D** 1 : 3
 - E** 3 : 1
 - F** 1 : 4
 - G** 4 : 1
 - H** 3 : 4
- 27** An alkene **X** with the molecular formula C_6H_{12} exists as a pair of *E/Z* (geometric) isomers.

Addition polymerisation of **X** forms a polymer in which each repeating unit has:

- exactly one alkyl group attached to each carbon atom of the two-carbon backbone;
- two alkyl groups that contain the same number of carbon atoms as each other.

What is the IUPAC name of alkene **X**?

- A** hex-1-ene
- B** hex-2-ene
- C** hex-3-ene
- D** 2-methylpent-2-ene
- E** 3-methylpent-2-ene
- F** 4-methylpent-2-ene
- G** 2,3-dimethylbut-2-ene